

File Edit Selection View Go Run Terminal Help python_test

EXPLORER

- PYTHON_TEST
 - custom_sketch.png
 - custom_vintage.png
 - DAY-01 Program Outp...
 - DAY-02 Program out...
 - DAY-03 Program Outp...
 - DAY-04 Program Outp...
 - DAY-05 Program Outp...
 - DAY-06 Program Outp...
 - DAY-07 Program Outp...
 - DAY-08 Program Outp...
 - DAY-09 Program and ...
 - day5_transformations....
 - day6_arithmetic_bitwis...
 - day7.mini.challenges.py
 - day8_color_spaces.py
 - day9_histograms.py
 - day10_smoothing_blu...
 - dissolve.mp4
 - drawing_basics.py
 - drawing.py
 - enhanced_image.png
 - equalization_comparis...
 - equalize.py
 - equalized_image.png
 - exercise_1_channel_art...
 - exercise_1_custom_pip...
 - exercise_1_histogram_...
 - exercise_2_color_picke...
 - exercise_2_compariso...
 - exercise_2_equalizatio...
 - exercise_3_3d_color_hi...
 - exercise_3_color_isolat...
 - exercise_4_color_enar...

day10_smoothing_blurring.py > ...

```
43 def add_salt_pepper_noise(img, prob=0.02):
44     output[coords[0], coords[1]] = 255
45
46     # Add pepper (black)
47     coords = [
48         np.random.randint(0, i, int(num_pepper))
49         for i in img.shape
50     ]
51     output[coords[0], coords[1]] = 0
52
53     return output
54
55 def add_gaussian_noise(img, mean=0, sigma=25):
56     """
57     Add Gaussian noise to image
58     """
59     noise = np.random.normal(mean, sigma, img.shape)
60     noisy = img + noise
61     return np.clip(noisy, 0, 255)
```

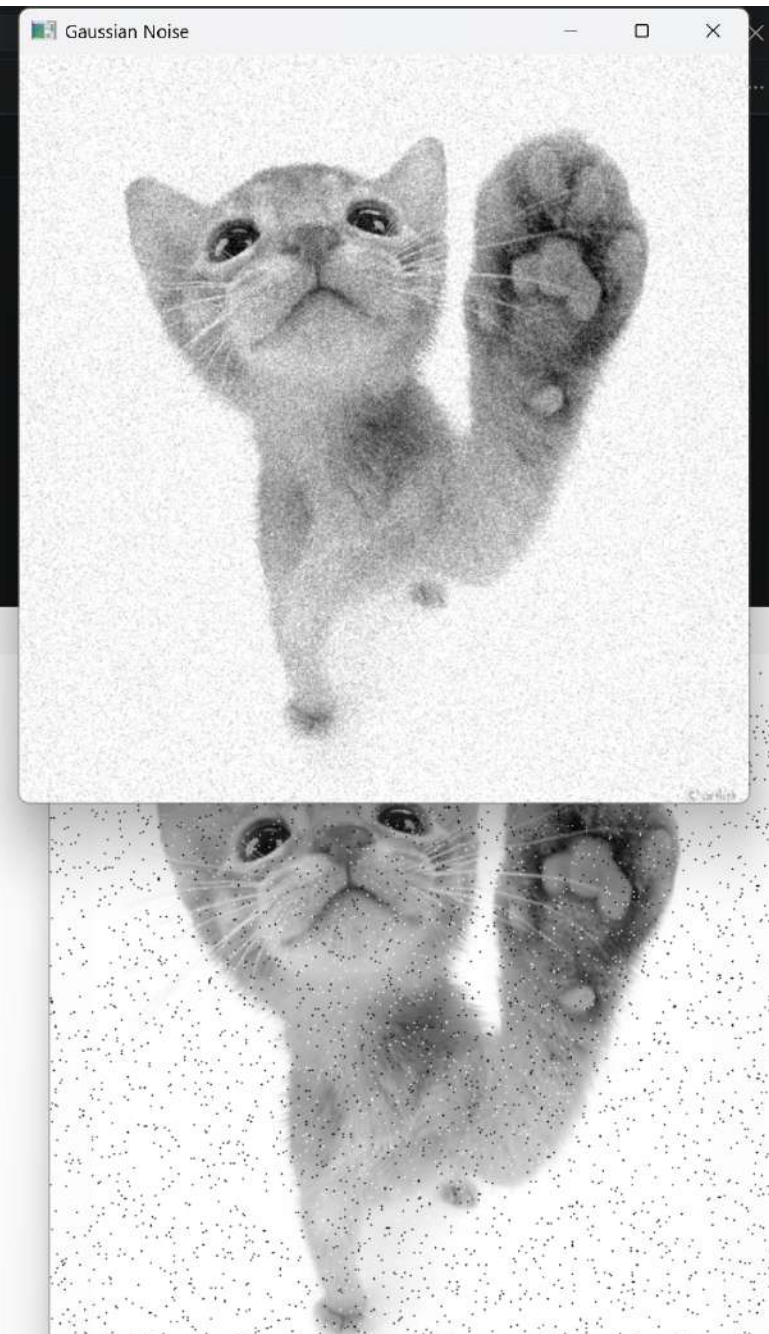
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

PS C:\Users\raghu\OneDrive\Desktop\python_test> python day10_smoothing_blurring.py

DAY 10: SMOOTHING AND BLURRING

[Section 1] Loading image and creating output image

Loaded: 474 x 486 pixels



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 - drawing.py
 - enhanced_image.png
 - equalization_comparis...
 - equalize.py
 - equalized_image.png
 - exercise_1_channel_art...
 - exercise_1_custom_pip...
 - exercise_1_histogram...
 - exercise_1_interactive...
 - exercise_2_color_picke...
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 - exercise_2_equalizatio...
 - exercise_3_3d_color_hi...
 - exercise_3_color_isolat
- OUTLINE
- TIMELINE

exercise_1_interactive_blur.py

```
def interactive_blur_comparison(image_path):  
    cv2.imshow(  
        "Blur Comparison",  
        comparison  
    )  
  
    key = cv2.waitKey(30) & 0xFF  
  
    if key == 27: # ESC  
        break  
  
    cv2.destroyAllWindows()  
  
# Run the interactive tool  
interactive_blur_comparison("test_image.png")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\raghu\OneDrive\Desktop\python_test> python exercise_1_intera
Interactive Blur Comparison Tool

Method: 0=Averaging, 1=Gaussian, 2=Median, 3=Bilateral
Kernel Size: 3, 5, 7, ..., 31 (odd numbers)
Sigma/Bilateral: for Gaussian sigma or bilateral sigma
Press ESC to exit
[]

Blur Comparison

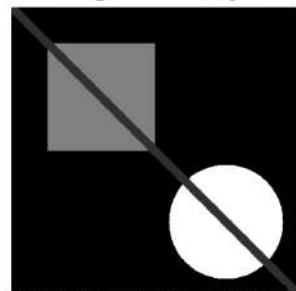
Kernel Size: 5 Method: 0

Sigma/Bilateral: 50

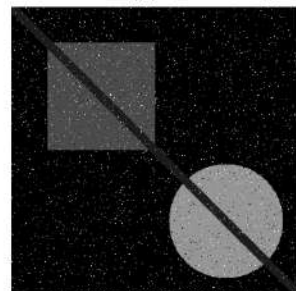
Method: Averaging
Kernel: 5x5

python

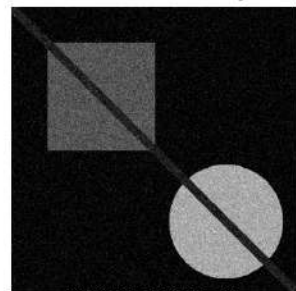
Original (Noisy)



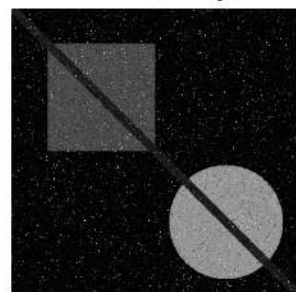
Salt & Pepper (Noisy)



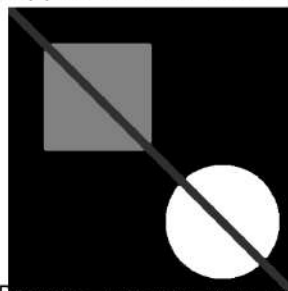
Gaussian (Noisy)



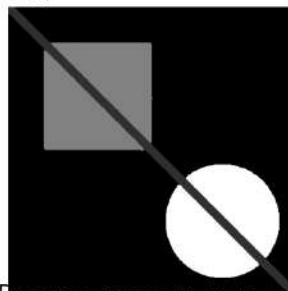
Mixed (Noisy)



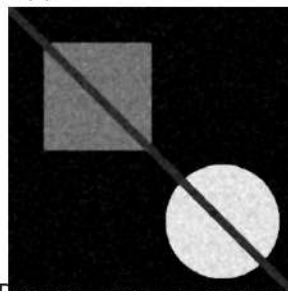
Salt-Pepper: Median Blur (best fo



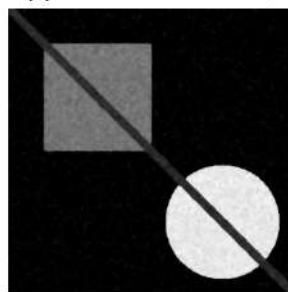
Salt-Pepper: Median Blur (best fo



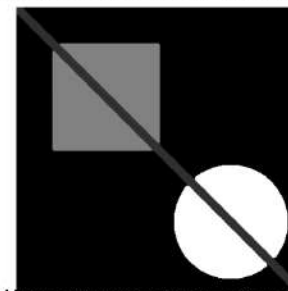
Salt-Pepper: Median Blur (best fo



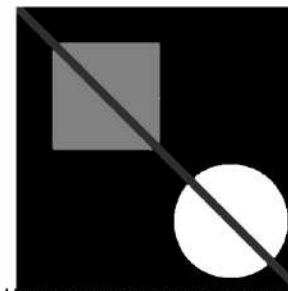
Salt-Pepper: Median Blur (best fo



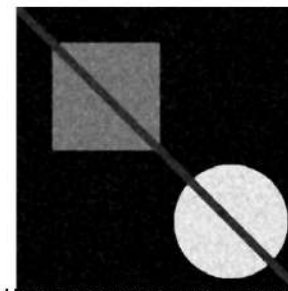
Auto: Median Blur (auto-de



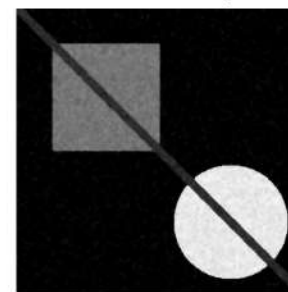
Auto: Median Blur (auto-de



Auto: Median Blur (auto-de



Auto: Median Blur (auto-de



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EXPLORER

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- exercise_3_color_isolat...
- exercise_3_custom_ker...
- exercise_4_color_spac...
- exercise_4_histogram...
- exercise_5_contrast_en...
- exercise1.brightness.c...
- exercise1.day3.py
- exercise1.house.py
- exercise1.py
- exercise1.thumbnail.py
- exercise2.day3.py
- exercise2.dissolve.py
- exercise2.panorama.py
- exercise2.py
- exercise2.target.py
- exercise3.animation.py
- exercise3.chessboard.py
- exercise3.day3.py

exercise_3_custom_kernel.py

```
1 # exercise_3_custom_kernel.py
2
3 import cv2
4 import numpy as np
5
6
7 def apply_custom_blur(image, kernel_type="sharpen", kernel_size=5):
8     """
9     Apply custom blur/sharpen kernels
10    """
11
12    if kernel_type == "sharpen":
13        kernel = np.array([
14            [-1, -1, -1],
15            [-1, 9, -1],
16            [-1, -1, -1]
17        ], dtype=np.float32)
18
19    elif kernel_type == "edge_detect":
20        kernel = np.array([
21            [-1, -1, -1],
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Kernel sum: 0.000

EMBOSS:
Kernel shape: (3, 3)
Kernel sum: 1.000

BOX_BLUR:
Kernel shape: (7, 7)
Kernel sum: 1.000

GAUSSIAN_CUSTOM:
Kernel shape: (7, 7)
Kernel sum: 1.000

MOTION_BLUR:
Kernel shape: (7, 7)
Kernel sum: 1.000

PS C:\Users\raghu\OneDrive\Desktop\python_test>

Ln 38, Col 1 Spaces: 4 UTF-8 CRLF {} Python 3,14.5 Python 3.14 (64-bit)

